

ETL Practical Task

Advanced Power BI – Data Engineering Simulation

Kato Kibirige Agathone

2300725696

23/U/25696/PSA

2026

Contents

1	Project Setup (Data Engineering Structure)	2
1.1	Folder Structure	2
1.2	Power BI Setup	2
2	Data Loading (Staging Layer)	2
2.1	Data Sources Loaded	3
2.2	Staging Configuration	3
3	Data Standardization Layer	3
3.1	Managers Transformation (stg_users → dim_managers_stg)	3
3.2	Orders Cleaning (stg_orders)	4
4	Data Integration Layer	5
4.1	Orders and Managers Join (fact_orders_enriched)	5
4.2	Northwind Integration (fact_northwind_sales)	6
5	Business Logic Transformation	7
5.1	Formatting and Type Conversions	7
5.2	Date Decomposition	8
5.3	Data Validation	9
6	Advanced Analytics Preparation	9
6.1	Customer Segmentation	9
6.2	Product Sales Ranking	10
6.3	Regional Performance Index	10
7	Final Data Warehouse Model	11
7.1	Star Schema Structure	12
7.2	Model Quality Rules Applied	12
8	Validation Tasks	12
8.1	Validation Checklist	13
8.2	Documentation Completion	13

1 Project Setup (Data Engineering Structure)

This section establishes the folder hierarchy and Power BI environment required for the ETL project.

1.1 Folder Structure

A dedicated desktop folder was created named: **ETL Advanced Kato Kibirige Agathone 2300725696**. The following sub-folders were created inside to represent each ETL layer:

- **01 Raw Data** – stores unmodified source files as received
- **02 Staging** – holds intermediate, unprocessed query outputs
- **03 Processed** – contains cleaned and transformed datasets
- **04 Output Reports** – stores final Power BI reports and exports

1.2 Power BI Setup

- Opened Power BI Desktop and saved the workbook as: **Advanced ETL Model.pbix**
- Enabled Power Query Editor profiling tools:
 - **Column Quality** – highlights valid, error, and empty cell percentages
 - **Column Distribution** – shows unique and distinct value counts per column
 - **Error Detection** – surfaces data type mismatches and transformation failures

2 Data Loading (Staging Layer)

Source data was loaded into Power Query without being pushed to the data model, keeping the staging layer separate from the model layer.

Column1	Column2
Region	Manager
Central	Chris
East	Erin
South	Sam
West	William

Figure 1: Staging process of loading data

2.1 Data Sources Loaded

- Supermarket Data (Excel/CSV format)
- Northwind Database (Microsoft Access .accdb format)

2.2 Staging Configuration

All loaded queries were configured with **Load to: Connection Only** to ensure no tables were pushed to the data model at this stage. Queries were renamed to standard staging conventions:

Staging Query Name	Source Table	Description
stg_orders	Sample Store Data	Raw order transactions
stg_customers	Northwind – Customers	Customer master records
stg_products	Northwind – Products	Product catalogue
stg_orderdetails	Northwind – Order Details	Line items per order
stg_users	Sample Store Data	User/manager records

3 Data Standardization Layer

This layer applies cleaning, renaming, and structural transformations to each staging query.

3.1 Managers Transformation (stg_users → dim_managers_stg)

- Renamed query from `stg_users` to `dim_managers_stg`
- Renamed columns for clarity:
 - Column1 → Region Code
 - Column2 → Manager FullName
- Split **Manager FullName** into two separate columns: **First Name** and **Last Name** (split by space delimiter)
- Created a **Surrogate Key** column using the following formula:

$$\text{Manager Key} = \text{HASH}(\text{Region Code} + \text{Manager FullName})$$

This ensures a unique, system-generated identifier for each manager record.

- Applied text standardization:
 - Trimmed leading and trailing spaces from all text columns
 - Removed special characters using Clean transformation
 - Standardized text casing (Title Case for names, UPPER for region codes)

Queries [17]

This preview may be up to 15 days old. Refresh

Table.RenameColumns(4*Added Index,{"Index", "Manager ID"})

	A ₁ Region Code	A ₂ First Name	A ₃ Last Name	A ₄ Manager Key	A ₅ Manager ID
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
	4 distinct, 4 unique	4 distinct, 4 unique	1 distinct, 0 unique		4 distinct, 4 unique
1	CENTRAL	Chris	N/A	M_CENTRAL_Chris_	1
2	EAST	Erin	N/A	M_EAST_Erin_	2
3	SOUTH	Sam	N/A	M_SOUTH_Sam_	3
4	WEST	William	N/A	M_WEST_William_	4

Figure 2: Transformation of managers table

Queries [17]

This preview may be up to 15 days old. Refresh

Table.RenameColumns(1.2*Discount)

	1.2 Discount
	Valid 100% Error 0% Empty 0%
	36 distinct, 36 unique
	0.03
	0.06
	0.07
	0.09
	0.05
	0.02
	0.01
	0.05
	0.08
	0.06
	0.09
	0.06
	0.01
	0.1

Figure 3: Disabling "Enable Load" to prevent table from appearing in data model

3.2 Orders Cleaning (stg_orders)

- Removed the following unnecessary columns to reduce data noise:
 - Row ID, Product Base Margin, Order Priority, Quantity Ordered New
- Filtered rows to include only the following five US states:
 - Arizona, California, Florida, New Jersey, New York
- Renamed columns for business-friendly naming:

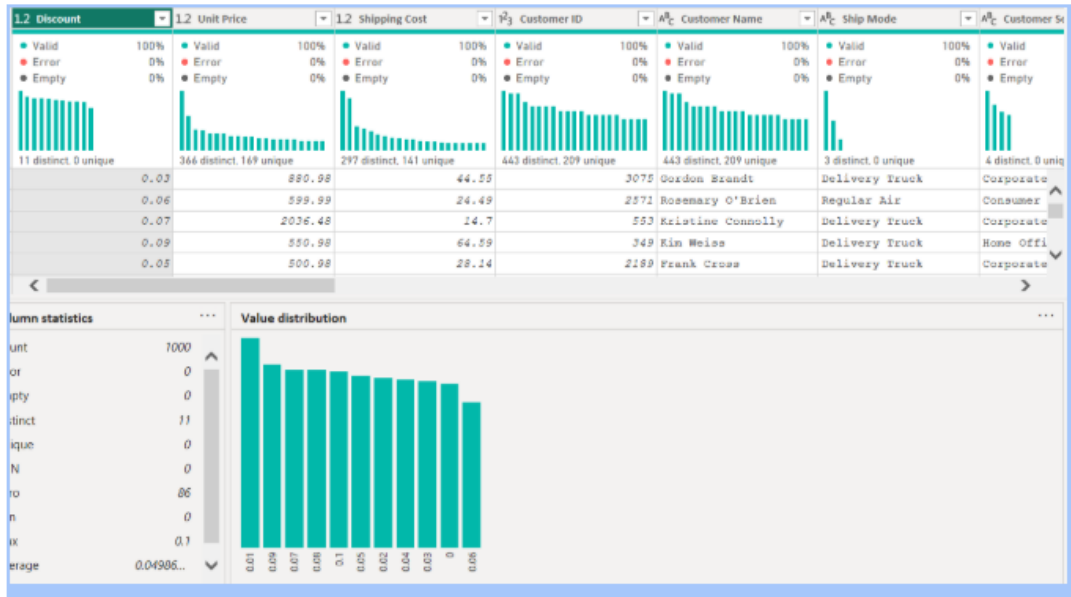


Figure 4: Data quality transformations

Original Column Name	Renamed To
State or Province	State
Postal Code	Zip Code
City	City Name

- Applied multi-column sort:
 - Primary Sort: State (Ascending)
 - Secondary Sort: Sales (Descending)

4 Data Integration Layer

Cleaned staging queries are merged in this layer to produce enriched fact tables.

4.1 Orders and Managers Join (fact_orders_enriched)

- Performed a **Left Outer Join** between `stg_orders` and `dim_managers_stg` on the **Region Code** column
- Expanded the following columns from the managers table after the join:
 - Manager Name, Region Code
- Created a conditional column: **Manager Performance Class**, based on Sales value:

Condition	Class Label
Sales > 10,000	High
Sales between 5,000 and 10,000	Medium
Sales < 5,000	Low

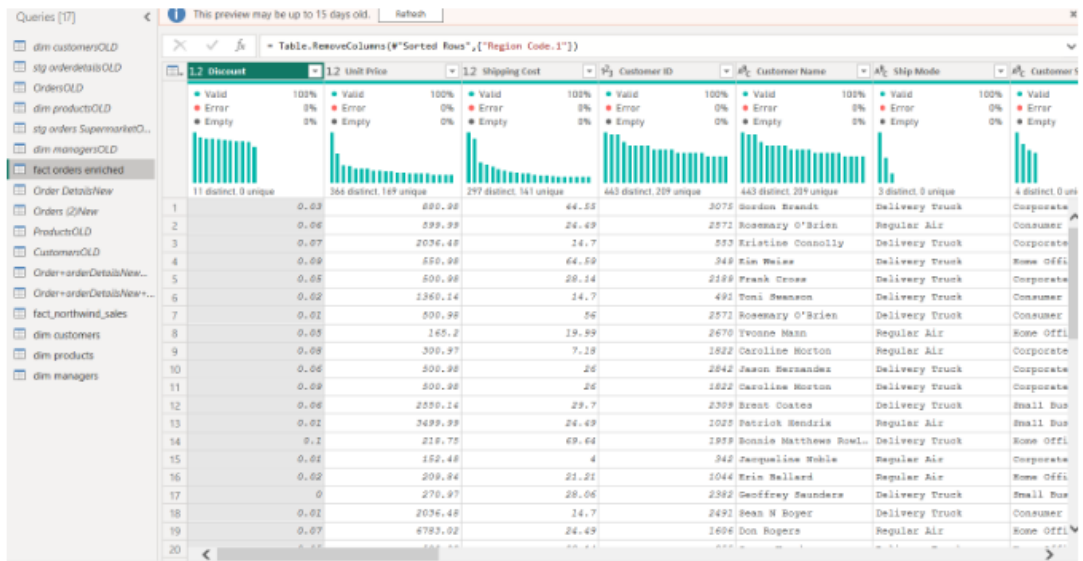


Figure 5: Fully created fact_orders_enriched table

- Filtered out all records where **Region Code = 'Central'**
- Applied sort by Manager Name (Ascending) then Sales (Descending)
- Saved and named the query: `fact_orders_enriched`

4.2 Northwind Integration (fact_northwind_sales)

- Performed a multi-step join across four Northwind tables:
 - Orders → Order Details (Join Key: OrderID)
 - Order Details → Products (Join Key: ProductID)
 - Orders → Customers (Join Key: CustomerID)
- Created three calculated measure columns:

Measure	Formula / Logic
Total Sales	Quantity × Unit Price
Discount Adjusted Sales	Total Sales × (1 – Discount)
Profit Proxy	Total Sales × 0.2

	Total Sales	Discount Adjusted Sales	Profit Proxy	Order Date - Year	Order Date - Quarter	Order Date - Month
1	2400	2400	280	2004	1	1
2	100	100	60	2004	1	1
3	300	300	60	2004	1	1
4	800	800	100	2004	1	1
5	20	20	7	2004	1	1
6	270	270	24	2004	1	1
7	600	600	100	2004	1	1
8	270	270	30	2004	1	1
9	100	100	30	2004	1	1
10	127.2	127.2	25.2	2004	1	2
11	1000	1000	380	2004	1	2
12	680	680	120	2004	1	2
13	13000	13000	2700	2004	1	3
14	1270	1270	200	2004	1	3
15	200	200	110.6	2004	1	3
16	13000	13000	2700	2004	1	3
17	200	200	20	2004	1	3
18	200	200	60	2004	1	3
19	80	80	18.4	2004	1	3
20	70	70	14	2004	1	3
21	149.7	149.7	29.9	2004	1	3

Figure 6: Fully created fact_northwind_sales table

Add a column that is computed from the other columns.

New column name:

Custom column formula:

Available columns:

- Order ID
- Employee ID
- Customer ID
- Order Date
- Shipped Date
- Shipper ID
- Ship Name

Learn about Power Query formulas

✓ No syntax errors have been detected.

Figure 7: Example of column creation in Power Query

- Removed all NULL records and orphan rows (records without matching keys across joined tables)
- Saved and named the query: `fact_northwind_sales`

5 Business Logic Transformation

This layer applies formatting, date extraction, and data validation across both fact tables.

5.1 Formatting and Type Conversions

- Converted **Discount** column from decimal (e.g. 0.10) to whole number percentage (e.g. 10), with 0 decimal places
- Formatted **Unit Price** as USD currency (e.g. \$12.50)

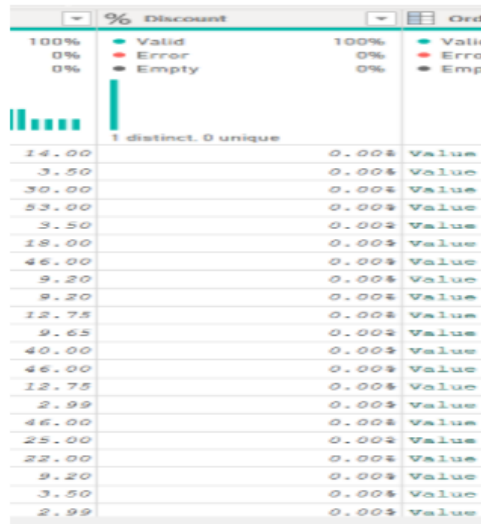


Figure 8: Discount column format changed to percentage

5.2 Date Decomposition

Extracted the following date components from the Order Date column:

- **Year** – numeric year extracted
- **Quarter** – Q1, Q2, Q3, or Q4 derived from month
- **Month** – month number (1–12) extracted
- **Weekday Name** column removed as it is not required for reporting

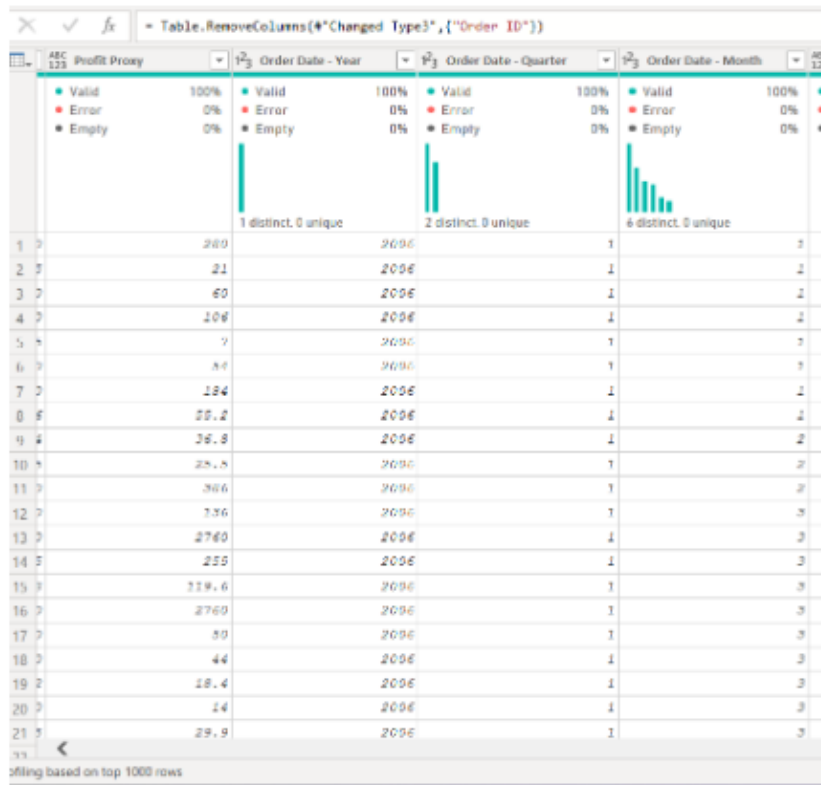


Figure 9: Date columns created for month, year, and weekday

5.3 Data Validation

- Verified no duplicate rows exist in either fact table
- Confirmed no NULL values appear in key columns (OrderID, CustomerID, ProductID, Manager Key)
- Validated all relationships reference correct and existing keys
- Ensured all columns have appropriate and consistent data types assigned

6 Advanced Analytics Preparation

Additional calculated columns were created to support dashboard-level analytics.

6.1 Customer Segmentation

A conditional column was added to classify customers based on their total Sales value:

Customer Segment
Low
Low
Low
Low
Low
Low
Low
Low
Low
Low
Low
Low
High

Figure 10: Customer segment column creation

Sales Range	Segment Label
High Sales threshold	High
Mid-range Sales	Medium
Low Sales threshold	Low

6.2 Product Sales Ranking

A rank column was added to Products, ordering them by total Sales value in descending order. This allows identification of top-performing products at the dashboard layer.

6.3 Regional Performance Index

A composite index column was calculated using a weighted formula:

$$\text{Regional Performance Index} = (\text{Sales} \times 0.5) + (\text{Profit} \times 0.3) + (\text{Discount Impact} \times 0.2)$$

This index enables comparison of regional contribution using a blended metric.

[illegible]

Figure 11: Regional Performance Index formula

7 Final Data Warehouse Model

The final model is structured as a **Star Schema**, linking fact tables to dimension tables through surrogate keys.

7.1 Star Schema Structure

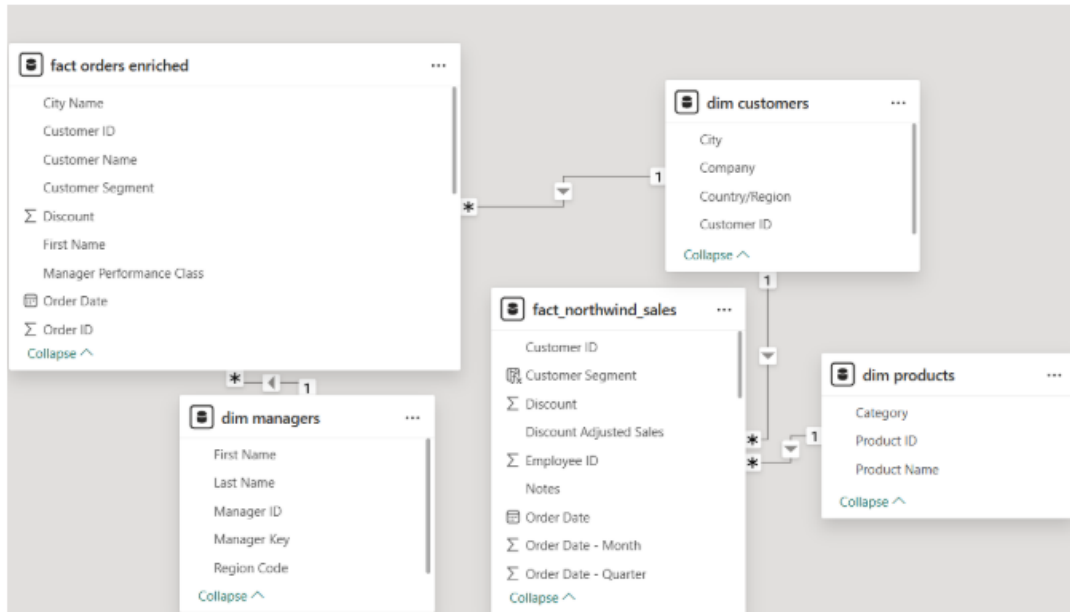


Figure 12: Star Schema data model

Table Name	Type	Description
fact_orders_enriched	Fact	Enriched sales orders with manager performance data
fact_northwind_sales	Fact	Northwind transactional sales with calculated measures
dim_managers	Dimension	Manager master data with surrogate keys
dim_products	Dimension	Product catalogue from Northwind
dim_customers	Dimension	Customer master records from Northwind

7.2 Model Quality Rules Applied

- No redundant columns retained across tables
- Relationships established only on surrogate or primary keys
- All keys are clean, non-NULL, and uniquely identify each record
- No many-to-many relationships exist in the model

8 Validation Tasks

A final validation sweep was performed across all fact and dimension tables before the dataset was released for dashboard design.

8.1 Validation Checklist

Validation Check	Status	Notes
No duplicate primary keys	Pass	Verified using Column Distribution in Power Query
No NULL values in fact tables	Pass	Error Detection enabled; nulls removed in step 4.2
Referential integrity maintained	Pass	All foreign keys match existing dimension keys
Correct data types assigned	Pass	Dates, numbers, text verified per column
No orphan records	Pass	Removed during Northwind join step
Weekday names removed	Pass	Column dropped post date extraction
Central region excluded	Pass	Filter applied in fact_orders_enriched

8.2 Documentation Completion

All transformation steps, from raw data ingestion through to the final validated data warehouse model, were documented in this report. The dataset was ready for Power BI dashboard design, including:

- Creation of calculated measures using DAX
- Design of interactive visuals (charts, slicers, maps)
- Publication and sharing via Power BI Service

End of Transformation Documentation